RAMCO INSTITUTE OF TECHNOLOGY
Approved by AICTE, New Delhi & Affiliated to Anna University
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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: II Semester B.E. EEE
Course Code & Title: EE3251 Electric Circuit Analysis
Name of the Faculty member (s): Mr. A. Arun Kumar

Innovative Practice Description

- **Unit / Topic:** Unit I / DC circuits – Kirchoff's Laws
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 2
- **Activity Chosen:** One minute paper
- **Justification:**

One-minute paper activity provides a conceptual bridge between successive class periods. Improve the quality of class discussion by having students write briefly about a concept what they understand in the class.

Time Allotted for the Activity: 02 minutes

- **Details of the Implementation:**

At the end of the class, students were asked to write about the topic discussed in the class. The students expressed the understood content and the content which were not clear in that particular topic. This activity shows whether the students can able to understand the specific topic and their involvement the particular class.

CO – PO / PSO mapping:

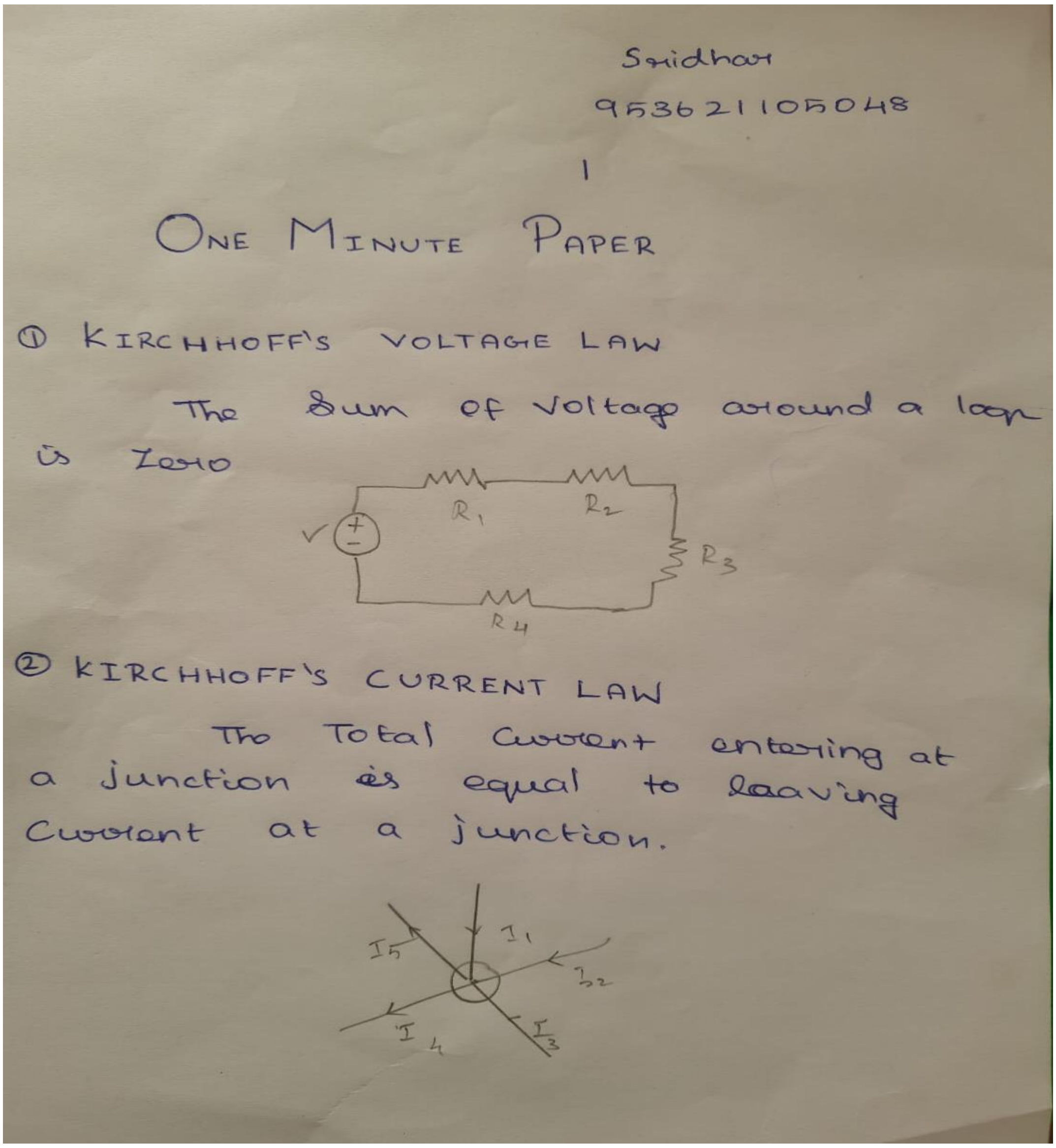
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	1	1	1	-	-	-	1	-	-	1	1	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as an individual

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
DC circuits – Kirchoff's Laws – One minute paper


- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
  - ❖ ***Benefit of the practice:***
    - ✓ Students can able to attend the question even in the questions are in indirect form.
    - ✓ Students can able to explain the concepts in examination without any confusion.
  - ❖ ***Challenges faced in implementation:***
    - ✓ Time utilization for conducting activity.

References:

1. William H. HaytJr, Jack E. Kemmerly and Steven M. Durbin, "Engineering Circuits Analysis", McGraw Hill publishers, 9thedition, New Delhi, 2020.
2. Charles K. Alexander, Mathew N.O. Sadiku, "Fundamentals of Electric Circuits", Second Edition, McGraw Hill, 2019.
3. Allan H. Robbins, Wilhelm C. Miller, "Circuit Analysis Theory and Practice", Cengage Learning India, 2013.

Signature of Faculty Member **HOD**

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Innovative Practice Description

❖ **Unit / Topic:** Unit II / Maximum Power Transfer theorem

- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 7
- **Activity Chosen:** Demonstration
- **Justification:**

  After teaching the concept of Maximum Power Transfer theorem, I thought of conducting demonstration for easy and better understanding of the concept.

  Time Allotted for the Activity: 10 minutes

- **Details of the Implementation:**
  Demonstration of maximum power transfer theorem using MATLAB/Simulink. The students can able to understand the concept easily

CO – PO / PSO mapping:

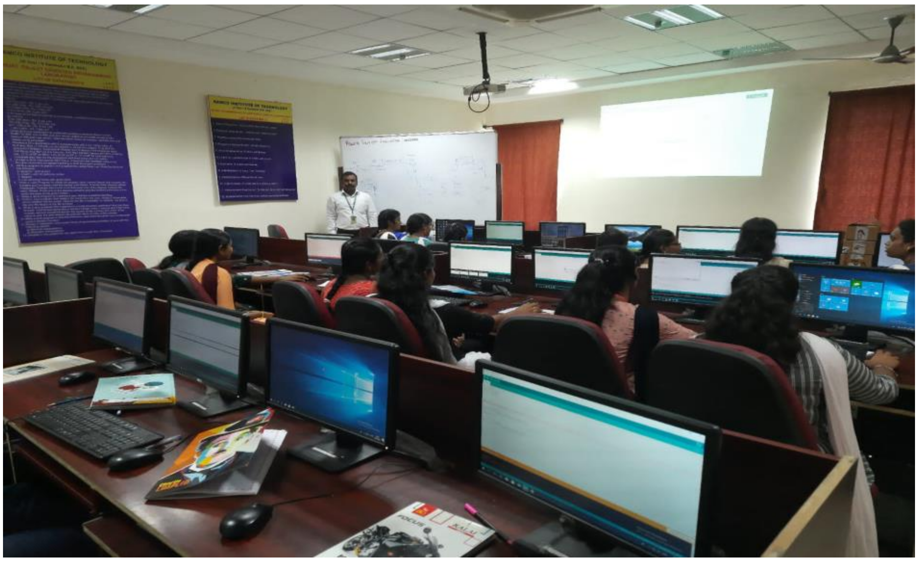
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	2	2	1	1	1	-	-	-	1	-	-	1	1	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO5**
	1
Justification for correlation	The students can Function effectively use the MATLAB software

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
Maximum Power Transfer theorem - Demonstration


- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

  - ***Benefit of the practice:***
    - ✓ Students can able to attend the question even in the questions are in indirect form.
    - ✓ Students can able to explain the concepts in examination without any confusion.

  - ***Challenges faced in implementation:***
    - ✓ Time utilization for conducting activity.

References:

1. William H. HaytJr, Jack E. Kemmerly and Steven M. Durbin, "Engineering Circuits Analysis", McGraw Hill publishers, 9thedition, New Delhi, 2020.
2. Charles K. Alexander, Mathew N.O. Sadiku, "Fundamentals of Electric Circuits", Second Edition, McGraw Hill, 2019.
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Innovative Practice Description

❖ **Unit / Topic:** Unit III / Transient response of RLC circuits

- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 10
- **Activity Chosen:** Think pair share
- **Justification:**
  - ✓ It helps students to think individually about a topic or answer to a question.
  - ✓ It teaches students to share ideas with classmates and builds oral communication skills.
  - ✓ It helps focus attention and engage students in comprehending the reading material.
- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

Think-Pair-Share innovative practice conducted for I year EEE students, after explained the concept of Transient response of RLC circuits. First, I asked the students to think about the comparison between Transient response of RLC circuits with step input and sine input for 2 minutes. Then I make them as a pair to discuss their neighbour's and asked the students to discuss about the same for 3 minutes. Finally, I asked the one of the team to explain the concept to whole class for further discussion. The students from group share their points and participated in the discussion for 10 minutes.

CO – PO / PSO mapping:

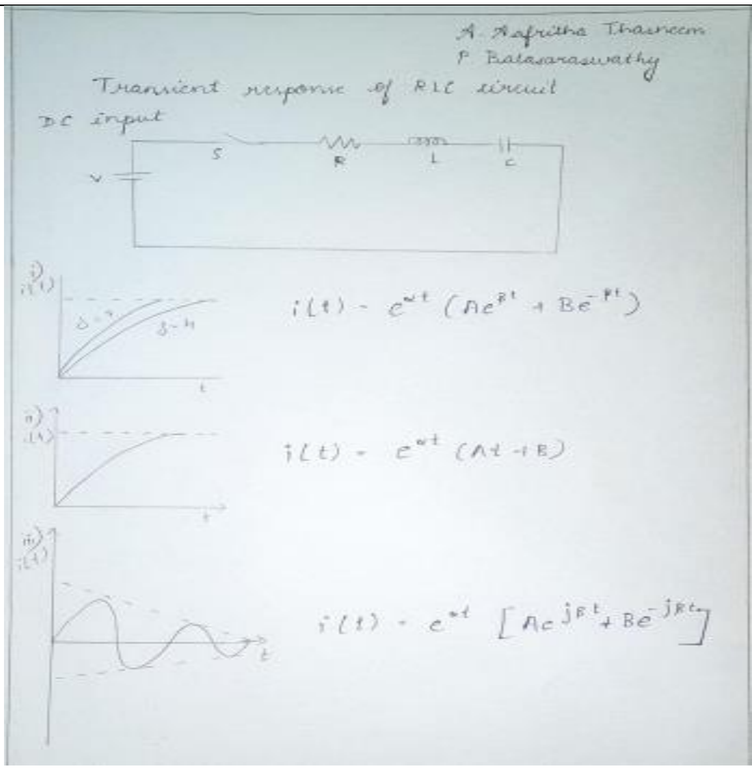
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	2	2	1	1	1	-	-	-	1	-	-	1	1	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO9**
	1
Justification for correlation	The students can Function effectively as an team

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
Transient response of RLC circuits – Think pair share


- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

  - ***Benefit of the practice:***

    Think-pair-sharing forces all students to attempt an initial response to the question, which they can then clarify and expand as they collaborate. It also gives them a chance to validate their ideas in a small group before mentioning them to the large group, which may help shy students feel more confident participating.

  - ***Challenges faced in implementation:***

    I planned the activity for 10 minutes. But in Class room it takes 20 minutes.

References:

1. William H. HaytJr, Jack E. Kemmerly and Steven M. Durbin, “Engineering Circuits Analysis”, McGraw Hill publishers, 9thedition, New Delhi, 2020.
2. Charles K. Alexander, Mathew N.O. Sadiku, “Fundamentals of Electric Circuits”, Second Edition, McGraw Hill, 2019.
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Innovative Practice Description

❖ **Unit / Topic:** Unit IV / Dot rule

- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 13
- **Activity Chosen:** Concept map

Concept mapping is a creative way to set goals, solve problems and design action plans. It quickly records ideas in a free-form way. When groups use concept mapping, the thoughts of each participant easily trigger ideas in others. This dynamic group interaction encourages breaking free of old patterns to uncover new and innovative approaches.

Time Allotted for the Activity: 05 minutes

- **Details of the Implementation:**

Concept mapping activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. When creating a concept map, a student frequently interacts with a textbook, notes from class, an instructor, classmate, or study group.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO4	2	2	1	1	1	-	-	-	1	-	-	1	1	-	-

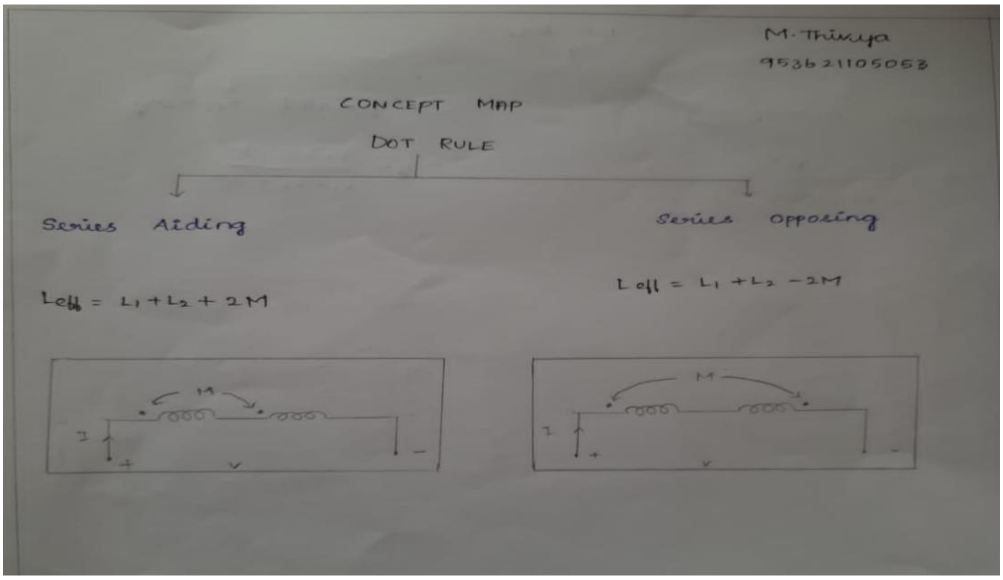
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO9**
	1
Justification for correlation	The students can Function effectively as an team

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
Dot rule – Concept map



- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***

    Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

  - ***Benefit of the practice:***

    The benefits of concept maps are a great way for students to make notes on all of the information they receive. It helps the students to note down only the most important information using key words, and then make connections between facts and ideas visually – keeping all of your topic thoughts together on one sheet. It made key note making easier to students, as it reduces pages of notes into one single side of paper. Also mind map made slow learners to remember the information more quickly.

  - ***Challenges faced in implementation:***

    I planned the activity for 05 minutes only. But in real scenario it takes 10 minutes to complete this activity.

References:

1. William H. HaytJr, Jack E. Kemmerly and Steven M. Durbin, “Engineering Circuits Analysis”, McGraw Hill publishers, 9thedition, New Delhi, 2020.
2. Charles K. Alexander, Mathew N.O. Sadiku, “Fundamentals of Electric Circuits”, Second Edition, McGraw Hill, 2019.
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Innovative Practice Description

❖ **Unit / Topic:** Unit V / Three phase power measurement

- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 17
- **Justification:**
  - ✓ The chosen topic will have the concepts related to design of various controllers for motor drive. Hence if the students are doing co-operative learning, then it is easier for them to understand the concepts. Sage and Scribe is one such cooperative learning activity.
  - ✓ The expected outcome is like a review process of learned concepts / points, Sage and Scribe is suitable activity.
- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**
  - ✓ The two students sitting adjacent are made to do the activity together. One student (Sage) will explain the concept for a while and other student (Scribe) will answer for the questions.
  - ✓ For the next question, the students will change their role. Likewise it will be repeated for all the questions.
  - ✓ The students will share the same piece of paper for doing this activity.
  - ✓ I clearly explained them that, since it is a cooperative activity both is responsible for the outcome.
  - ✓ At the end of the lecture hour, the students are made to explain their answers to the class. I have given the correct answers for the questions with explanation.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO5	2	2	1	1	1	-	-	-	1	-	-	1	1	-	-

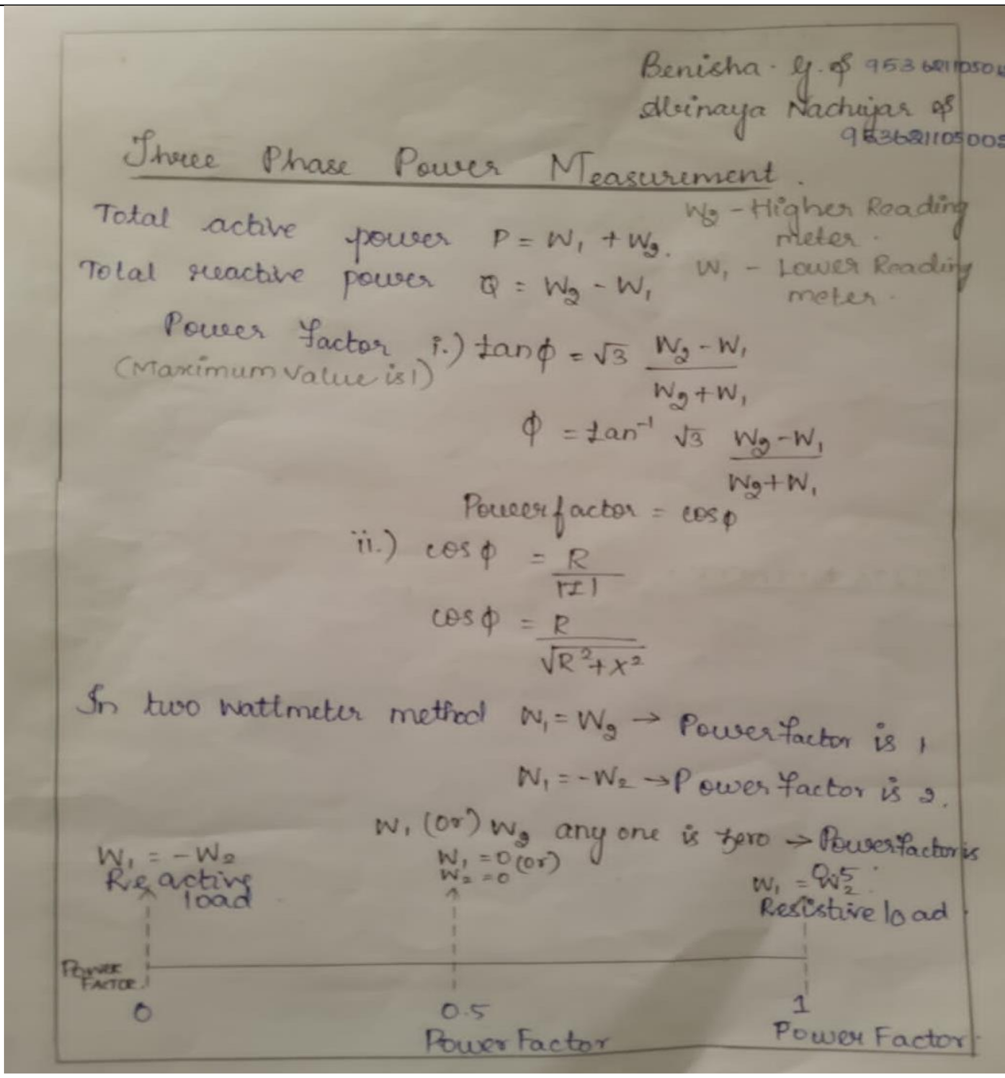
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO9**
	1
Justification for correlation	The students can Function effectively as an team

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
Three phase power measurement – Sage and Scribe



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Abinaya Nachiyar 953621105003

Three Phase Power Measurement.

Total active power $P = W_1 + W_2$.

W_2 – Higher Reading meter.

W_1 – Lower Reading meter.

Total reactive power $Q = W_2 - W_1$

Power factor (Maximum value is 1) i.) $\tan\phi = \sqrt{3}\,\frac{W_2 - W_1}{W_2 + W_1}$

$\phi = \tan^{-1} \sqrt{3}\,\frac{W_2 - W_1}{W_2 + W_1}$

Power factor $= \cos\phi$

ii.) $\cos\phi = \frac{R}{|Z|}$

$\cos\phi = \frac{R}{\sqrt{R^2 + X^2}}$

In two wattmeter method $W_1 = W_2 \rightarrow$ Power factor is 1

$W_1 = -W_2 \rightarrow$ Power factor is 0.

W_1 (or) W_2 any one is zero $\rightarrow$ Power factor is 0.5

- **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

✓ The students have liked the session and they felt comfortable in doing this activity, since they are interacting with their neighbor. They did the work easily.

✓ They conveyed that more such activity will help them in understanding the concepts clearly and easily.

❖ **Benefit of the practice:**

✓ The students have understood the concepts clearly and it has been evident from the answers they have written on the answer paper.

✓ The students have presented the answers to the class. It cleared any doubt in understanding the important points in the chosen topic.

❖ **Challenges faced in implementation:**

✓ In few groups, both the students were slow learners; hence they find it very difficult to answer the questions. I helped them in answering the questions by providing them with simple hints. In the next time, while doing this activity, I should properly pair the students

References:

1. William H. HaytJr, Jack E. Kemmerly and Steven M. Durbin, "Engineering Circuits Analysis", McGraw Hill publishers, 9thedition, New Delhi, 2020.
2. Charles K. Alexander, Mathew N.O. Sadiku, "Fundamentals of Electric Circuits", Second Edition, McGraw Hill, 2019.
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